Chapter 10. Regression Diagnostics

Hat matrix and leverage, residuals, deletion formulas, influence measures

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10.1 What can go wrong

Fitting $\mathbf{y} = X\beta + \epsilon$, $\epsilon \sim N(\mathbf{0}, \sigma^2 I)$, by $\hat{\beta} = (X'X)^{-1}X'\mathbf{y}$ is justified only if the model is adequate. **Regression diagnostics** look for:

(1) violations of the error assumptions (normality, independence, equal variance), studied through residuals;

(2) a wrong functional form, remedied by transforming y or the x_j (Chapters 5 and 11);

(3) near-singularity of $X'X$ (multicollinearity, Chapter 15);

(4) **high-leverage points**: observations whose $\mathbf{x}_i$ is far from the bulk of the design;

(5) **outliers**: observations whose y_i is far from its fitted value;

(6) **influential observations**: observations whose deletion changes the fit substantially.

Types (4)–(6) need not coincide: a high-leverage point can lie on the line, and a central outlier may barely move the fit.

10.2 The hat matrix and leverage

Motivation. The hat-matrix diagonal measures how far a point lies in x-space, flagging leverage before looking at y.

The fitted values are $\hat{\mathbf{y}} = H\mathbf{y}$ with $H = X(X'X)^{-1}X' = (h_{ij})$, where $h_{ij} = \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_j$ and $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$ is the i-th row of X.

Theorem 10.1 (Properties of the hat matrix)

(a) H is symmetric and idempotent with $\operatorname{rank} H = \operatorname{tr} H = \sum_i h_{ii} = p + 1$.

(b) $h_{ii} = \sum_j h_{ij}^2 = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$; hence $0 \le h_{ii} \le 1$ and $|h_{ij}| \le \frac{1}{2}$ for $i \neq j$.

(c) $HX = X$. If the model has an intercept, $H\mathbf{1} = \mathbf{1}$, i.e. every row of H sums to 1, and $h_{ii} \ge 1/n$.

(d) With an intercept, $h_{ii} = \frac{1}{n} + (\mathbf{x}_i^* - \bar{\mathbf{x}})'(\tilde{X}'\tilde{X})^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$, where $\mathbf{x}_i^* = (x_{i1}, \ldots, x_{ip})'$ and $\tilde{X}$ is the centred $n \times p$ matrix of regressors. For $p = 1$, $h_{ii} = \frac{1}{n} + \frac{(x_i - \bar{x})^2}{S_{xx}}$.

Proof. (a) $H' = H$, $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$, and $\operatorname{tr} H = \operatorname{tr}\{(X'X)^{-1}X'X\} = p + 1$.

(b) The (i,i) entry of $H = HH'$ is $\sum_j h_{ij}^2$. Since it is at least h_{ii}^2, $h_{ii} \ge h_{ii}^2$, so $0 \le h_{ii} \le 1$. For $j \neq i$, $h_{ij}^2 \le h_{ii} - h_{ii}^2 \le \frac{1}{4}$.

(c) $HX = X$ is direct; the intercept column gives $H\mathbf{1} = \mathbf{1}$. For $h_{ii} \ge 1/n$ see (d).

(d) With centred regressors $\tilde{X}$ orthogonal to $\mathbf{1}$, $H = \frac{1}{n}\mathbf{1}\mathbf{1}' + \tilde{X}(\tilde{X}'\tilde{X})^{-1}\tilde{X}'$; take diagonals (the second term is nonnegative). For $p = 1$, $\tilde{X}'\tilde{X} = S_{xx}$. ∎

As $\hat{y}_i = \sum_j h_{ij} y_j$, $h_{ii} = \partial \hat{y}_i / \partial y_i$ is the **leverage** of observation i; by (d) it grows with the Mahalanobis distance $M(i) = (\mathbf{x}_i^* - \bar{\mathbf{x}})'\{\tilde{X}'\tilde{X}/(n-1)\}^{-1}(\mathbf{x}_i^* - \bar{\mathbf{x}})$:

$$M(i) = (n-1)\Big(h_{ii} - \frac{1}{n}\Big).$$

The average leverage is $\bar{h} = (p+1)/n$, and a common rule flags $h_{ii} > 2\bar{h}$ (or $3\bar{h}$) as high leverage.

10.3 Residuals

Motivation. Scaling residuals by their standard errors, with the point deleted, gives t-comparable outlier statistics.

Theorem 10.2

$\mathbf{e} = (I-H)\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2(I-H))$. So $\operatorname{Var} e_i = \sigma^2(1-h_{ii})$, $\operatorname{Cov}(e_i, e_j) = -\sigma^2 h_{ij}$, and $\operatorname{Corr}(e_i, e_j) = -h_{ij}/\sqrt{(1-h_{ii})(1-h_{jj})}$.

High-leverage points have small residual variance: the fit is pulled towards them. To compare residuals on a common scale:

Definition (Scaled residuals)

With $s^2 = MSE = \mathbf{e}'\mathbf{e}/(n-p-1)$ and $s^2_{(i)}$ the MSE computed without observation i:

- the **standardized** (internally studentized) residual is $r_i = \dfrac{e_i}{s\sqrt{1-h_{ii}}}$;
- the **studentized** (externally studentized) residual is $r_i^* = \dfrac{e_i}{s_{(i)}\sqrt{1-h_{ii}}}$.

r_i is not t: e_i and s are dependent (large $|e_i|$ inflates s). Deleting i removes this. Let $X_{(i)}$, $\mathbf{y}_{(i)}$ omit row i, with estimator $\hat\beta_{(i)}$ and prediction $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)}$.

Theorem 10.3 (Deletion formulas)

(a) $\hat\beta_{(i)} = \hat\beta - \dfrac{(X'X)^{-1}\mathbf{x}_i e_i}{1-h_{ii}}$.

(b) $y_i - \tilde y_i = \dfrac{e_i}{1-h_{ii}}$ (the **deleted** or PRESS residual).

(c) $(n-p-2)s^2_{(i)} = (n-p-1)s^2 - \dfrac{e_i^2}{1-h_{ii}}$.

Proof [Supplement]. (a) $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. With $A = X'X$, the Sherman–Morrison formula gives

$$(A - \mathbf{x}_i\mathbf{x}_i')^{-1} = A^{-1} + \frac{A^{-1}\mathbf{x}_i\mathbf{x}_i'A^{-1}}{1-h_{ii}}.$$

Multiplying by $X'\mathbf{y} - \mathbf{x}_i y_i$ and using $A^{-1}X'\mathbf{y} = \hat\beta$, $\mathbf{x}_i'\hat\beta = \hat y_i$, $\mathbf{x}_i'A^{-1}\mathbf{x}_i = h_{ii}$:

$$\hat\beta_{(i)} = \hat\beta - A^{-1}\mathbf{x}_i y_i + \frac{A^{-1}\mathbf{x}_i(\hat y_i - h_{ii}y_i)}{1-h_{ii}} = \hat\beta + \frac{A^{-1}\mathbf{x}_i\{\hat y_i - h_{ii}y_i - (1-h_{ii})y_i\}}{1-h_{ii}} = \hat\beta - \frac{A^{-1}\mathbf{x}_i e_i}{1-h_{ii}}.$$

(b) $\tilde y_i = \mathbf{x}_i'\hat\beta_{(i)} = \hat y_i - h_{ii}e_i/(1-h_{ii})$, so $y_i - \tilde y_i = e_i + h_{ii}e_i/(1-h_{ii}) = e_i/(1-h_{ii})$.

(c) For $j \neq i$, the deleted residual is $y_j - \mathbf{x}_j'\hat\beta_{(i)} = e_j + h_{ji}e_i/(1-h_{ii})$. Summing squares over $j \neq i$ and using $\sum_j e_j h_{ji} = 0$ (since $H\mathbf{e} = \mathbf{0}$) and $\sum_j h_{ji}^2 = h_{ii}$:

$$SSE_{(i)} = \sum_{j\neq i} e_j^2 + \frac{2e_i}{1-h_{ii}}\sum_{j\neq i} h_{ji}e_j + \frac{e_i^2}{(1-h_{ii})^2}\sum_{j\neq i} h_{ji}^2.$$

Here $\sum_{j\neq i} e_j^2 = SSE - e_i^2$, $\sum_{j\neq i} h_{ji}e_j = -h_{ii}e_i$, and $\sum_{j\neq i} h_{ji}^2 = h_{ii} - h_{ii}^2$. So

$$SSE_{(i)} = SSE - e_i^2 - \frac{2h_{ii}e_i^2}{1-h_{ii}} + \frac{h_{ii}e_i^2}{1-h_{ii}} = SSE - \frac{e_i^2}{1-h_{ii}}.$$

■

Theorem 10.4 (Distribution of the studentized residual)

(a) $r_i^* = \dfrac{y_i - \tilde{y}_i}{s_{(i)}\sqrt{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i}} \sim t(n-p-2)$.

(b) $r_i^* = r_i\sqrt{\dfrac{n-p-2}{n-p-1-r_i^2}}$, a monotone function of r_i.

Proof. (a) y_i is independent of $\hat{\beta}_{(i)}$ and $s_{(i)}^2$, which are computed from the other $n-1$ observations; so $y_i - \tilde{y}_i \sim N(0, \sigma^2\{1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i\})$, independent of $(n-p-2)s_{(i)}^2/\sigma^2 \sim \chi^2(n-p-2)$. By Sherman–Morrison, $1 + \mathbf{x}_i'(X_{(i)}'X_{(i)})^{-1}\mathbf{x}_i = 1 + h_{ii} + h_{ii}^2/(1-h_{ii}) = 1/(1-h_{ii})$, and with Theorem 10.3(b) the ratio equals $e_i/\{s_{(i)}\sqrt{1-h_{ii}}\}$.

(b) Substitute Theorem 10.3(c), $s_{(i)}^2 = s^2(n-p-1-r_i^2)/(n-p-2)$. ■

So an observation is flagged as an outlier when $|r_i^*| > t_{\alpha/2}(n-p-2)$. Since n tests are performed, a Bonferroni level α/n is safer.

10.4 Influence measures

An observation is influential if deleting it changes the fitted model a lot. The measures below combine the residual and the leverage.

Theorem 10.5 (Influence measures)

(a) **DFFITS**: $\mathrm{DFFITS}_{(i)} = \dfrac{\hat{y}_i - \tilde{y}_{i(i)}}{s_{(i)}\sqrt{h_{ii}}} = \left(\dfrac{h_{ii}}{1-h_{ii}}\right)^{1/2} r_i^*$, where $\tilde{y}_{i(i)} = \mathbf{x}_i'\hat{\beta}_{(i)}$.

(b) **Cook's distance**: $D_{(i)} = \dfrac{(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)})}{(p+1)s^2} = \dfrac{r_i^2}{p+1}\cdot\dfrac{h_{ii}}{1-h_{ii}}$.

(c) **COVRATIO**: $\mathrm{COVRATIO}_{(i)} = \dfrac{\det\{s_{(i)}^2(X_{(i)}'X_{(i)})^{-1}\}}{\det\{s^2(X'X)^{-1}\}} = \dfrac{1}{(1-h_{ii})}\left(\dfrac{n-p-2+r_i^{*2}}{n-p-1}\right)^{-(p+1)}$.

(d) **Andrews–Pregibon**: with $Z = (X, \mathbf{y})$, $\mathrm{AP}_{(i)} = \dfrac{\det(Z_{(i)}'Z_{(i)})}{\det(Z'Z)} = 1 - h_{ii} - \dfrac{e_i^2}{SSE}$.

Proof [Supplement]. (a) By Theorem 10.3(a), $\hat{y}_i - \tilde{y}_{i(i)} = \mathbf{x}_i'(\hat{\beta} - \hat{\beta}_{(i)}) = h_{ii}e_i/(1-h_{ii})$; divide by $s_{(i)}\sqrt{h_{ii}}$.

(b) $(\hat{\beta} - \hat{\beta}_{(i)})'X'X(\hat{\beta} - \hat{\beta}_{(i)}) = \frac{e_i^2}{(1-h_{ii})^2}\mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i = \frac{h_{ii}e_i^2}{(1-h_{ii})^2}$; divide by $(p+1)s^2$.

(c) By the matrix determinant lemma,

$$\det(X_{(i)}'X_{(i)}) = \det(X'X)\,(1-h_{ii}),$$

so the ratio is $(s^2_{(i)}/s^2)^{p+1}/(1-h_{ii})$. By Theorem 10.3(c), $s^2_{(i)}/s^2 = (n-p-1-r_i^2)/(n-p-2)$, and by Theorem 10.4(b), $n-p-2+r_i^{*2} = (n-p-2)(n-p-1)/(n-p-1-r_i^2)$. Hence $s^2_{(i)}/s^2 = (n-p-1)/(n-p-2+r_i^{*2})$, which gives the formula.

(d) By the determinant lemma, $\det(Z'_{(i)}Z_{(i)})/\det(Z'Z) = 1-\mathbf{z}_i'(Z'Z)^{-1}\mathbf{z}_i$, $\mathbf{z}_i' = (\mathbf{x}_i', y_i)$: the leverage in the regression on Z, which equals $h_{ii} + e_i^2/SSE$ (partitioned inverse). ■

Rules of thumb (Belsley, Kuh and Welsch; Cook):

- $|\text{DFFITS}_{(i)}| > 2\sqrt{(p+1)/n}$;
- $D_{(i)} > F_{0.50}(p+1, n-p-1)$ (roughly 1): deleting i moves $\hat\beta$ outside the 50% confidence ellipsoid of Theorem 8.2;
- $|\text{COVRATIO}_{(i)} - 1| > 3(p+1)/n$;
- small $\text{AP}_{(i)}$.

Every measure grows with both $|r_i|$ and h_{ii}: influence is "outlyingness × leverage".

Example 10.1 (Two kinds of unusual point)

For $n = 21$ children (age at first word x, aptitude y), $\hat y = 109.87 - 1.13x$. Observation 18 ($x = 42$) has $h_{18,18} = 0.65 > 3\bar h = 0.29$ and a small residual ($r^* = -0.85$), yet the largest $D_{(18)} = 0.68$ and $\text{DFFITS}_{(18)} = -1.16$: influential. Observation 19 has $r^*_{19} = 3.61 > t_{0.025}(18) = 2.10$: an outlier, but with ordinary leverage ($h = 0.053$), so Cook's distance is 0.22. Check the outlier's measurement; assess dependence on the high-leverage point.

What to do. Check unusual points for recording errors; a genuine one may signal a wrong model. Robust regression or fits with and without the point beat silent deletion. Single-deletion diagnostics can miss groups of points that mask each other.

10.5 Selected exercises

Motivation. A point on the fitted surface, leverage one, and why standardized residuals are not t.

Exercise 10.1 (Deleting a point on the fitted surface)

Suppose $y_i = \hat y_i$ and $X_{(i)}$ has full column rank. Show from the normal equations that deleting i leaves the LSE unchanged, and find $s^2_{(i)}$ in terms of s^2.

Solution.

Step 1. $X'_{(i)}X_{(i)} = X'X - \mathbf{x}_i\mathbf{x}_i'$ and $X'_{(i)}\mathbf{y}_{(i)} = X'\mathbf{y} - \mathbf{x}_i y_i$. Hence, for the full-data estimator $\hat\beta$,

$$X'_{(i)}(\mathbf{y}_{(i)} - X_{(i)}\hat\beta) = X'(\mathbf{y} - X\hat\beta) - \mathbf{x}_i(y_i - \mathbf{x}_i'\hat\beta) = \mathbf{0} - \mathbf{x}_i e_i = \mathbf{0},$$

since $X'\mathbf{e} = \mathbf{0}$ and $e_i = 0$.

Step 2. So $\hat\beta$ solves the normal equations of the reduced data. As $X'_{(i)}X_{(i)}$ is nonsingular, the solution is unique: $\hat\beta_{(i)} = \hat\beta$. (This is the case $e_i = 0$ of Theorem 10.3(a).)

Step 3. The remaining residuals are unchanged, so $SSE_{(i)} = \sum_{j\neq i} e_j^2 = SSE$, now with $n-p-2$ degrees of freedom: $s^2_{(i)} = \frac{n-p-1}{n-p-2}s^2$, which is larger than s^2 when $SSE > 0$.

Such a point has zero Cook's distance however large its leverage, illustrating that leverage alone does not make a point influential. ■

Exercise 10.2 (Leverage equal to one)

Suppose $h_{ii} = 1$.
(a) Show that $h_{ij} = 0$ for $j \neq i$, and that $\hat{y}_i = y_i$ and $e_i = 0$ for every response vector $\mathbf{y}$.
(b) Show that $X'_{(i)}X_{(i)}$ is singular, so that β cannot be estimated without observation i; conversely, if $X_{(i)}$ has full column rank, then $h_{ii} < 1$.
(c) In simple regression with $x = (0, \dots, 0, 1)$ (with $n \geq 2$), compute h_{nn}.

Solution [Supplement].

(a) By Theorem 10.1(b), $h_{ii} = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2$, i.e. $1 = 1 + \sum_{j \neq i} h_{ij}^2$, so every $h_{ij} = 0$ $(j \neq i)$. Then $\hat{y}_i = \sum_j h_{ij} y_j = y_i$ and $e_i = 0$, whatever $\mathbf{y}$ is.

(b) By the matrix determinant lemma (as in the proof of Theorem 10.5(c)),

$$\det(X'_{(i)}X_{(i)}) = \det(X'X - \mathbf{x}_i\mathbf{x}_i') = \det(X'X)\,\{1 - \mathbf{x}_i'(X'X)^{-1}\mathbf{x}_i\} = \det(X'X)(1 - h_{ii}).$$

So $h_{ii} = 1$ makes $X'_{(i)}X_{(i)}$ singular, i.e. $X_{(i)}$ rank deficient. Conversely, if $X_{(i)}$ has full column rank, the determinant is positive, so $h_{ii} \neq 1$, and $h_{ii} < 1$ by Theorem 10.1(b).

(c) $\bar{x} = 1/n$ and $S_{xx} = (n-1)/n^2 + (1 - 1/n)^2 = (n-1)/n$. By Theorem 10.1(d),

$$h_{nn} = \frac{1}{n} + \frac{(1 - 1/n)^2}{(n-1)/n} = \frac{1}{n} + \frac{n-1}{n} = 1.$$

The others all have $x = 0$, so the point at $x = 1$ alone determines the slope, and the line passes through it. ■

Exercise 10.3 (Standardized residuals are bounded)

Assume $h_{ii} < 1$.
(a) Show that $r_i^2 \leq n - p - 1$. Conclude that r_i does not have a t distribution.
(b) Show that the map $r_i \mapsto r_i^*$ of Theorem 10.4(b) is a strictly increasing bijection from $(-\sqrt{n-p-1}, \sqrt{n-p-1})$ onto $\mathbb{R}$.

Solution [Supplement].

(a) By Theorem 10.3(c), $0 \leq (n-p-2)s_{(i)}^2 = (n-p-1)s^2 - e_i^2/(1-h_{ii})$. Hence $e_i^2/\{s^2(1-h_{ii})\} \leq n-p-1$, i.e. $r_i^2 \leq n-p-1$, with equality iff $SSE_{(i)} = 0$. A t variable takes arbitrarily large values with positive probability, so r_i cannot be t-distributed.

(b) On $|r| < \sqrt{n-p-1}$ let $f(r) = r\sqrt{(n-p-2)/(n-p-1-r^2)}$, continuous and odd. For $0 \leq r_1 < r_2$ both positive factors increase, so $f(r_1) < f(r_2)$; by oddness f is strictly increasing, with $f \to \pm\infty$ at $\pm\sqrt{n-p-1}$, hence onto $\mathbb{R}$ (IVT). So r_i^* and r_i rank alike, but only r_i^* has the unbounded $t(n-p-2)$ law. ■